Normalization Report - Blood Bank Management System

Objective

Design a scalable, normalized SQL Server database for donor management, blood inventory tracking, hospital blood requests, donation history management, and reporting support.

1NF

Repeating donor and blood request details are separated into individual tables. Donor, hospital, blood inventory, donation, and request data are stored as atomic values in separate tables.

2NF

Partial dependency is removed. Donor details depend only on DonorID, hospital details depend only on HospitalID, inventory details depend only on InventoryID, and donation/request attributes depend completely on their primary keys.

3NF

Transitive dependency is removed. Hospital information is maintained only in the Hospital table, donor information is maintained only in the Donor table, and blood stock information is maintained only in the BloodInventory table to avoid redundancy.

BCNF

Every determinant is a candidate key or primary key. Unique attributes such as donor phone number and inventory identifiers are properly constrained to maintain consistency.

Integrity and Validation

The schema uses primary keys, foreign keys, NOT NULL, UNIQUE, and CHECK constraints to maintain data integrity and validation.

Reporting Support

Views, indexes, functions, joins, subqueries, and aggregation queries support blood stock reporting, donor activity tracking, hospital request monitoring, and inventory management.

Entity Relationship Justification

Relationship	Cardinality	Justification
Donor - BloodDonation	1:M	One donor can donate blood multiple times.
Hospital - BloodRequest	1:M	One hospital can make multiple blood requests.
BloodInventory - BloodRequest	1:M	One inventory can fulfill multiple requests.